# PGCP-AC August 26
# Database Assignment No-3

1. Write a query to calculate the total salary of all employees.
Answer:

```
mysql> SELECT SUM(SALARY) AS TOTAL_SAL FROM EMP;
+------------+
| TOTAL_SAL |
+------------+
| 509300.00 |
+------------+
1 row in set (0.00 sec)
```

2. Write a query to find the average salary of employees in each department using GROUP BY.
Answer:

```
mysql> SELECT D.DNAME , ROUND(AVG(E.SALARY)) AS AVG_SALARY FROM DEPT D JOIN EMP E ON D.DEPT_
ID = E.DEPT_ID GROUP BY D.DEPT_ID;
+-----------+------------+
| DNAME     | AVG_SALARY |
+-----------+------------+
| HR        |      51250 |
| Finance   |      51500 |
| Marketing |      38500 |
| IT        |      62150 |
+-----------+------------+
4 rows in set (0.00 sec)
```

3. Write a query to count the total number of employees in each department.
Answer:

```
mysql> SELECT D.DNAME,COUNT(E.EMP_ID) AS TOTAL_EMP FROM DEPT D JOIN EMP E ON D.DEPT_ID = E.D
EPT_ID GROUP BY D.DEPT_ID;
+-----------+-----------+
| DNAME     | TOTAL_EMP |
+-----------+-----------+
| IT        |         2 |
| HR        |         4 |
| Finance   |         2 |
| Marketing |         2 |
+-----------+-----------+
4 rows in set (0.00 sec)
```

4. Write a query to display departments having more than 2 employees using HAVING clause.
Answer:

```
mysql> SELECT D.DNAME,COUNT(E.EMP_ID) FROM DEPT D JOIN EMP E ON D.DEPT_ID = E.DEPT_ID GROUP
BY D.DEPT_ID HAVING COUNT(E.EMP_ID)>2;
+-------+-----------------+
| DNAME | COUNT(E.EMP_ID) |
+-------+-----------------+
| HR    |               4 |
+-------+-----------------+
1 row in set (0.00 sec)
```

5. Write a query to list distinct department locations from the Dept table.
Answer:

```
mysql> SELECT DISTINCT LOCATION FROM DEPT;
+----------+
| LOCATION |
+----------+
| Mumbai   |
| Pune     |
| Nashik   |
| Thane    |
| Panvel   |
+----------+
5 rows in set (0.00 sec)
```

6. Write a query to display the highest salary among all employees. Write a query to display all total number of employee in table.
Answer:

```
mysql> SELECT MAX(SALARY) FROM EMP;
+-------------+
| MAX(SALARY) |
+-------------+
|    65000.00 |
+-------------+
1 row in set (0.00 sec)

mysql> SELECT COUNT(*) FROM EMP;
+----------+
| COUNT(*) |
+----------+
|       10 |
+----------+
1 row in set (0.00 sec)
```

7. Write a query to find employees whose name starts with 'A' using LIKE operator.
Answer:

```
mysql> SELECT NAME FROM  EMP WHERE NAME LIKE 'A%';
+--------+
| NAME   |
+--------+
| Amit   |
| Anjali |
+--------+
2 rows in set (0.00 sec)
```

8. Write a query to find employees whose name ends with 'n' using LIKE operator.
Answer:

```
mysql> SELECT NAME FROM EMP WHERE NAME LIKE '%N';
+-------+
| NAME  |
+-------+
| Rohan |
| Karan |
+-------+
2 rows in set (0.00 sec)
```

9. Write a query to find employees whose name contains 'ra' using LIKE operator.
Answer:

```
mysql> SELECT NAME FROM EMP WHERE NAME LIKE '%RA%';
+-------+
| NAME  |
+-------+
| Rahul |
| Karan |
+-------+
2 rows in set (0.00 sec)
```

10. Write a query to display all employees sorted by their Salary in descending order.
Answer:

```
mysql> SELECT * FROM EMP ORDER BY SALARY DESC;
+--------+--------+----------+---------+---------+------+
| EMP_ID | NAME   | SALARY   | JOINING | DEPT_ID | NOTE |
+--------+--------+----------+---------+---------+------+
|    108 | Pooja  | 65000.00 |    2019 |       3 | NULL |
|    110 | Anjali | 63800.00 |    2020 |       1 | NULL |
|    106 | Neha   | 60500.00 |    2021 |       1 | NULL |
|    104 | Sneha  | 60000.00 |    2020 |       2 | NULL |
|    102 | Priya  | 52000.00 |    2021 |       2 | NULL |
|    107 | Karan  | 48000.00 |    2023 |       2 | NULL |
|    101 | Rahul  | 45000.00 |    2022 |       2 | NULL |
|    105 | Rohan  | 42000.00 |    2022 |       4 | NULL |
|    103 | Amit   | 38000.00 |    2023 |       3 | NULL |
|    109 | Vikas  | 35000.00 |    2024 |       4 | NULL |
+--------+--------+----------+---------+---------+------+
10 rows in set (0.00 sec)
```

11. Write a query to display all employees sorted by DeptID ascending and then Salary descending.

Answer:

```
mysql> SELECT * FROM EMP ORDER BY DEPT_ID ASC , SALARY DESC;
+--------+--------+----------+---------+---------+------+
| EMP_ID | NAME   | SALARY   | JOINING | DEPT_ID | NOTE |
+--------+--------+----------+---------+---------+------+
|    110 | Anjali | 63800.00 |    2020 |       1 | NULL |
|    106 | Neha   | 60500.00 |    2021 |       1 | NULL |
|    104 | Sneha  | 60000.00 |    2020 |       2 | NULL |
|    102 | Priya  | 52000.00 |    2021 |       2 | NULL |
|    107 | Karan  | 48000.00 |    2023 |       2 | NULL |
|    101 | Rahul  | 45000.00 |    2022 |       2 | NULL |
|    108 | Pooja  | 65000.00 |    2019 |       3 | NULL |
|    103 | Amit   | 38000.00 |    2023 |       3 | NULL |
|    105 | Rohan  | 42000.00 |    2022 |       4 | NULL |
|    109 | Vikas  | 35000.00 |    2024 |       4 | NULL |
+--------+--------+----------+---------+---------+------+
10 rows in set (0.00 sec)
```

12. Write a query to find employees whose salary is between 30,000 and 60,000.

Answer:

```
mysql> SELECT * FROM EMP WHERE SALARY BETWEEN 30000 AND 60000;
+--------+-------+----------+---------+---------+------+
| EMP_ID | NAME  | SALARY   | JOINING | DEPT_ID | NOTE |
+--------+-------+----------+---------+---------+------+
|    101 | Rahul | 45000.00 |    2022 |       2 | NULL |
|    102 | Priya | 52000.00 |    2021 |       2 | NULL |
|    103 | Amit  | 38000.00 |    2023 |       3 | NULL |
|    104 | Sneha | 60000.00 |    2020 |       2 | NULL |
|    105 | Rohan | 42000.00 |    2022 |       4 | NULL |
|    107 | Karan | 48000.00 |    2023 |       2 | NULL |
|    109 | Vikas | 35000.00 |    2024 |       4 | NULL |
+--------+-------+----------+---------+---------+------+
7 rows in set (0.00 sec)
```

13. Write a query to display all employees whose DeptID is in (1, 2, 3).

Answer:

```
mysql> SELECT NAME FROM EMP WHERE DEPT_ID IN (1,2,3);
+--------+
| NAME   |
+--------+
| Rahul  |
| Priya  |
| Amit   |
| Sneha  |
| Neha   |
| Karan  |
| Pooja  |
| Anjali |
+--------+
8 rows in set (0.00 sec)
```

14. Write a query to display Min salary of employee .
Answer:

```
mysql> SELECT MIN(SALARY) FROM EMP;
+-------------+
| MIN(SALARY) |
+-------------+
|    35000.00 |
+-------------+
1 row in set (0.00 sec)
```

15. Write a query to display employees whose JoiningDate is between '2020' and '2021'.
Answer:

```
mysql> SELECT NAME FROM EMP WHERE JOINING BETWEEN 2020 AND 2021;
+--------+
| NAME   |
+--------+
| Priya  |
| Sneha  |
| Neha   |
| Anjali |
+--------+
4 rows in set (0.00 sec)
```

16. Write a query to display employees whose Salary is NULL.
Answer:

```
mysql> SELECT NAME FROM EMP WHERE SALARY IS NULL;
+------+
| NAME |
+------+
| Amit |
+------+
1 row in set (0.00 sec)
```

17. Write a query to display employees whose Salary is NOT NULL.
Answer:

```
mysql> SELECT NAME FROM EMP WHERE SALARY IS NOT NULL;
```

18. Write a query to calculate the total salary per department, but only for departments where total salary is greater than 1,00,000 (use HAVING).
Answer:

```
mysql> SELECT D.DNAME , SUM(E.SALARY) FROM DEPT D JOIN EMP E ON D.DEPT_ID = E.DEPT_ID GROUP BY D.
DEPT_ID HAVING SUM(E.SALARY) >100000;
+-------+---------------+
| DNAME | SUM(E.SALARY) |
+-------+---------------+
| HR    |     205000.00 |
| IT    |     124300.00 |
+-------+---------------+
2 rows in set (0.00 sec)
```

19. Write a query to display all distinct employee names.
Answer:

```
mysql> SELECT DISTINCT NAME FROM EMP;
```

20. Write a query to count the number of departments having the same location.
Answer:

```
mysql> SELECT LOCATION FROM DEPT GROUP BY LOCATION HAVING COUNT(*)>1;
+----------+
| LOCATION |
+----------+
| Mumbai   |
+----------+
1 row in set (0.00 sec)
```